

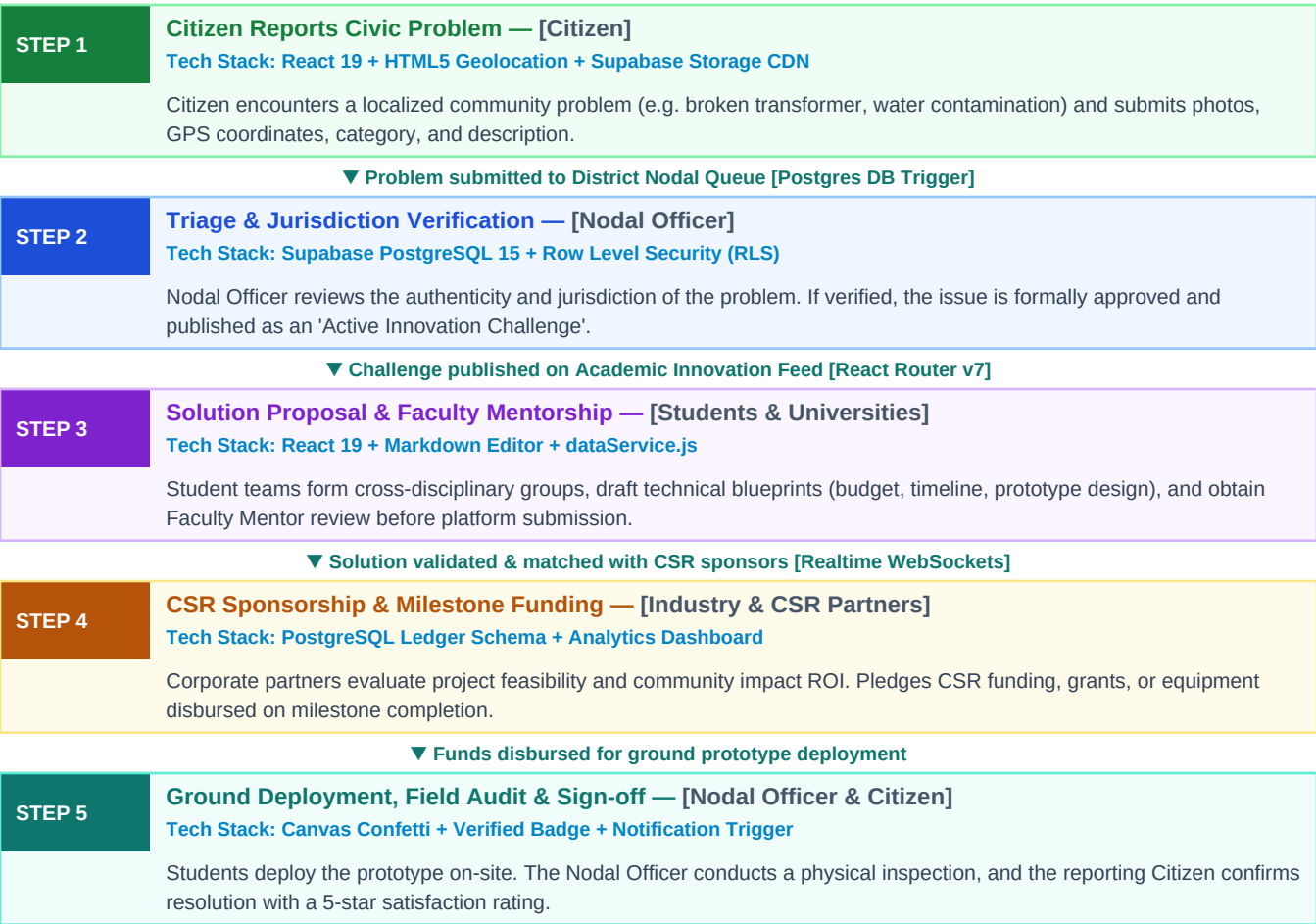
1. Complete Technology Stack Matrix

Layer	Technologies & Frameworks Used	Architectural Role & Functionality
Frontend Framework	React 19, Vite 8, React Router v7	Ultra-fast SPA rendering, component modularity, client-side routing, optimistic state updates.
Styling & UI	Tailwind CSS v3, PostCSS, Lucide Icons	Modern glassmorphism, responsive grid layouts, high-contrast accessible civic design tokens.
Auth & RBAC	Supabase Auth, OAuth 2.0 (Google, GitHub)	Passwordless secure OAuth login, JWT sessions, first-time role onboarding modal, RBAC.
Database & Persistence	Supabase (PostgreSQL 15), Row Level Security	Relational schema (profiles, problems, solutions, projects, messages), real-time sync.
Backend Gateway	Vercel Serverless (/api/chat.js), Vite Middleware	Zero-exposure environment secrets handling, CORS, rate limiting, and multi-model AI routing.
AI Engine	Google Gemini API (@google/genai, gemini-2.5-pro/flash)	25 strict civic domains, out-of-domain rejection, step-by-step problem-solving guidance.
Micro-Animations	Canvas Confetti (canvas-confetti), Status Badges	Visual milestone rewards and interactive feedback upon problem approval and resolution.

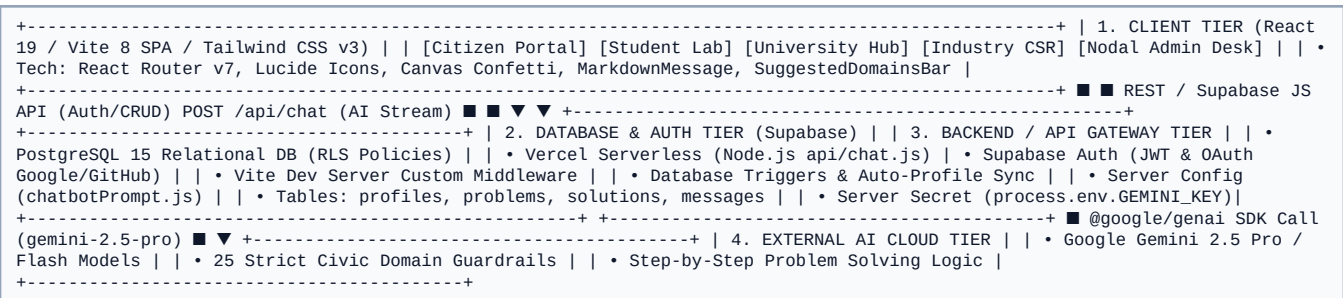
2. How the Backend Works (In Simple Language)

- 1. Secure Gateway & Key Isolation (Vercel Serverless / Node.js):** When a user asks a question or submits a challenge, the frontend calls POST /api/chat. The backend decrypts server environment variables (process.env.GEMINI_API_KEY), guaranteeing that secret credentials never reach the browser.
- 2. Dual-Engine Resilience (@google/genai + Local Knowledge Base):** The backend attempts live reasoning via Google Gemini 2.5 Pro / Flash. If quota or network limits occur, it seamlessly falls back to the **Intelligent Civic Knowledge Engine**, returning verified step-by-step answers without failing.
- 3. Role-Based Data Security (Supabase PostgreSQL 15 + RLS):** Row Level Security policies and database triggers guarantee that citizens only edit their own problems, students collaborate securely, and Nodal Officers sign off on genuine civic projects.

3. Flow Chart 1: Master Civic Problem-Solving Lifecycle (With Tech Stacks)



4. Flow Chart 2: System Architecture Data Pipeline (With Tech Stacks)



5. Flow Chart 3: Authentication & Role Onboarding (With Tech Stacks)

[illegible]

6. Flow Chart 4: Samadhan AI 25-Domain Decision Engine (With Tech Stacks)

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[ User sends message or clicks Chip ] <#### [ Tech: MessagingPage.jsx + SuggestedDomainsBar.jsx ] ■ ▼ [ Frontend sanitizes input & slices history to last 10 messages ] <#### [ Tech: aiChatService.js ] ■ ▼ [ POST /api/chat JSON Payload ] <#### [ Tech: Fetch API HTTP Stream ] ■ ▼ [ Backend Gateway checks process.env.GEMINI_API_KEY ] <#### [ Tech: Vercel Serverless / Node.js Runtime ] #### API Key Present >>> [ Initialize Google Gen AI SDK ] <#### [ Tech: @google/genai SDK v2.20.0 ] ■ ■ ■ ■ ▼ ■ [ Call Gemini 2.5 Pro / Flash ] <#### [ Tech: ai.models.generateContent() ] ■ ■ ■ ■ Success >>> [ Return JSON reply & status: success ] ■ ■ ■ ■ 404/429/Timeout >>> [ Failover to Local Knowledge Engine ] ■ ■ ■ ■ No API Key ■ ■ ■ ■ ■ ■ ■ ■ [ Run Intelligent Local Knowledge Engine ] <#### [ Tech: 25-Domain Heuristic Engine ] #### Is In-Domain? ■ ■ ■ ■ > [ Return Step-by-Step Civic Steps ] ■ ■ ■ ■ Is Out-of-Domain? >> [ Return Official Canned Refusal ] ■ ▼ [ Frontend receives JSON reply & renders UI ] <#### [ Tech: MarkdownMessage.jsx (RegExp Tokenizer) ]
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Specification Summary: This document provides the complete frontend, backend, database, and AI technology stack flowcharts for Samadhan.Connect. Engineered for state-wide citizen empowerment, robust security, and seamless civic collaboration.